

An Overview of the Kähler Minimal Model Program

Kähler MMP

Yi Li

1 Overview

The aim of this note is to give a brief survey on recent developments in the Kähler minimal model program. This note is intended as an overview of what is currently known about the Kähler minimal model program, and briefly sketches some of the key ideas that appear along the way.

Throughout, the reader is assumed to be familiar with the projective minimal model program at the level of the cone, contraction and base point free theorems, and with the transcendental objects that replace divisors in the Kähler setting: Bott–Chern cohomology classes and the positivity cones, quasi-plurisubharmonic functions, closed positive currents and Lelong numbers, and generalized pairs. These are set up in the companion notes of this series, to which the relevant sections below refer. Results here are stated rather than proved; where a proof is sketched, it is marked as an idea of proof and the details are deferred to the note of this series devoted to that result.

Contents

1	Overview	1
2	Cone Theorems	2
3	Rational Curves on Kähler Varieties and Ou’s BDPP Theorem for Compact Kähler Manifolds	2
4	Contraction Theorems	3
5	Transcendental Base Point Freeness	4
6	Existence of Minimal Models, Good Minimal Models, and Canonical Models	4
7	Kähler Criterion and Projectivity Criterion	6

8	Canonical Bundle Formulas	7
9	Abundance Conjecture for Kähler Pairs	7
10	Several Applications of Kähler minimal model program	7

2 Cone Theorems

The original proof of the cone theorem relies on Mori's bend-and-break technique. Unfortunately, this technique remains unavailable for Kähler manifolds. Using the recent resolution of the BDPP conjecture [Ou25], the cone theorem for Kähler varieties has nonetheless been established.

Theorem 2.1 ([HP24, Theorem 0.5]). *Let X be a compact $\mathbb{Q}$ -factorial Kähler variety of dimension n such that $(X, B + \beta)$ is generalized klt and $K_X + B + \beta_X$ is pseudo-effective. Then the following hold:*

(1) (Cone theorem) *There are at most countably many rational curves $\{\Gamma_i\}_{i \in I}$ such that*

$$\overline{\text{NA}}(X) = \overline{\text{NA}}(X)_{K_X + B + \beta_X \geq 0} + \sum_{i \in I} \mathbb{R}^+ [\Gamma_i].$$

(2) (Length of extremal rays) $0 < -(K_X + B + \beta_X) \cdot \Gamma_i \leq 2n$.

(3) (Finiteness) *If $B + \beta_X$ or $K_X + B + \beta_X$ is big, then I is finite.*

💡 Idea of proof. A key observation is that the maximal-dimension component Z of the null locus $\text{Null}(\alpha)$ is covered by rational curves.

Let us give a remark on the history of cone theorems in the Kähler minimal model program: In [HP16], Höring–Petersen developed a Mori bend-and-break theorem for Kähler threefolds, using which they proved the cone theorem for Kähler threefolds with terminal singularities. Later, [CHP16, DO24] use a similar idea to generalize it to the cone theorem for Kähler threefold dlt pairs. Motivated by [CH20], [HP24] completes the proof of the cone theorem for Kähler klt pairs in any dimension.

3 Rational Curves on Kähler Varieties and Ou's BDPP Theorem for Compact Kähler Manifolds

Rational curves play a central role in birational geometry, which is also the case for the Kähler minimal model program.

Recently, Ou [Ou25] proved the BDPP conjecture for compact Kähler manifolds.

Theorem 3.1 ([Ou25, Theorem 1]). *Let X be a compact Kähler manifold. Then K_X is not pseudo-effective if and only if X is uniruled.*

Mori bend-and-break is a well-established theory for projective varieties. However, to date it is known only for Kähler threefolds.

Theorem 3.2 ([HP16, Corollary 5.7]). *Let X be a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at most terminal singularities such that K_X is pseudo-effective. Let $S_1, \dots, S_k$ be components of the Zariski decomposition of K_X . Let*

$$b := \max \{1, -K_X \cdot Z \mid Z \text{ a curve s.t. } Z \subset S_{j, \text{sing}} \text{ or } Z \subset S_j \cap S_{j'}\},$$

and set

$$d := \max\{3, b\}$$

If $C \subset X$ is a curve such that $-K_X \cdot C > d$, then we have

$$[C] = [C_1] + [C_2]$$

with C_1 and C_2 effective 1-cycles (with integer coefficients, need not be irreducible) on X .

4 Contraction Theorems

Hacon–Xie [HLX26a] completes the proof of contraction theorem for Kähler minimal model program. The proof of this result is quite technical; we will devote a great deal of effort in this series to explaining it clearly.

Theorem 4.1 ([HLX26a, Theorem 1.6]). *Let $(X, B + \beta)$ be a n -dimensional compact Kähler generalized pair such that $B + \beta_X$ is modified big and $\alpha = K_X + B + \beta_X$ is nef and NQC. Assume that the contraction defined by $[\alpha]_N$ exists and is Moishezon, where $N = \text{Nklt}(X, B + \beta)$ is the closed subspace defined by the multiplier ideal $\mathcal{J}(X, B + \beta)$. Then the contraction defined by $[\alpha]$ exists and is Moishezon.*

Note that for a gklt pair the multiplier ideal is trivial, so $N = \emptyset$ and the hypothesis on $[\alpha]_N$ is vacuous; the theorem is stated for a possibly non-gklt pair because the induction in [HLX26a] runs over log canonical centres, on which gklt fails.

Historically, for Kähler threefolds Höring–Petersen [HP16] claimed the result for $\mathbb{Q}$ -factorial compact Kähler threefolds with terminal singularities. Das–Hacon [DH25, DH24a] gave a complete proof for Kähler threefolds with $\mathbb{Q}$ -factorial dlt singularities (see [DH24a, Theorem 1.2]).

When a flipping contraction exists, Das–Hacon proved the following result on the existence of flips for generalized klt Kähler pairs in any dimension.

Theorem 4.2 ([DH24a, Theorem 5.12]). *Let $(X, B + \beta)$ be a $\mathbb{Q}$ -factorial gdlt pair, where X is a compact analytic variety belonging to Fujiki’s class C , and $f : X \rightarrow Z$ a flipping contraction. Then the flip of f exists.*

💡 Idea of proof. The idea of the proof is to use the local existence of log canonical models; since these are unique, they glue to the flip.

5 Transcendental Base Point Freeness

The transcendental base-point-freeness is conjectured by Tosatti [Tos18], which has been solved recently by Hacon–Xie [HLX26a].

Theorem 5.1 ([HLX26a, Theorem 1.4]). *Let $(X, B + \beta)$ be a compact, n -dimensional, Kähler, gklt pair such that $\alpha = K_X + B + \beta_X$ is nef, and $B + \beta_X$ is modified big. Then α is semiample, more precisely, there exists a Moishezon contraction $f : X \rightarrow Y$ such that*

- (1) Y is a compact Kähler space, and
- (2) $\alpha \equiv f^* \gamma$ for some Kähler class γ on Y .

Moreover, if X is strongly $\mathbb{Q}$ -factorial, then f is projective.

Historically, for projective pairs the base-point-free theorem holds; see [DH24b, Theorem 3.5].

6 Existence of Minimal Models, Good Minimal Models, and Canonical Models

In this section, we will talk about the existence of minimal model in Kähler MMP.

6.1 Existence of Minimal Models for Kähler Threefolds

For Kähler threefolds, the result is relatively complete ([DH24a], using Shokurov’s difficulty function for termination of the MMP).

Theorem 6.1 ([DH24a, Theorem 1.4]). *Let (X, B) be a dlt pair where X is a $\mathbb{Q}$ -factorial compact Kähler threefold. If $K_X + B$ is pseudo-effective, then there exists a finite sequence of*

flips and divisorial contractions

$$\phi : X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_n$$

*such that $K_{X_n} + \phi_*B$ is nef.*

Later, Das–Hacon–Yáñez proved the result for generalized Kähler threefolds.

Theorem 6.2 ([DHY23, Theorem 1.2]). *Let $(X, B + \beta)$ be a generalized compact klt Kähler threefold pair.*

(1) *If $K_X + B + \beta_X$ is big, then $(X, B + \beta)$ has a log terminal model $f : X \dashrightarrow X^m$ and a unique log canonical model $g : X^m \rightarrow X^c$.*

(2) *If $K_X + B + \beta_X$ is pseudo-effective and β_X is big, then $K_X + B + \beta_X$ has a log terminal model $f : X \dashrightarrow X^m$, and there is a contraction $g : X^m \rightarrow Z$ such that $f_*(K_X + B + \beta_X) \equiv g^*\omega_Z$, where ω_Z is a Kähler form on Z .*

6.2 Existence of Minimal Models for Kähler Fourfold Effective Pairs

For effective Kähler fourfold dlt pairs, the existence of minimal models was proved by Das–Hacon–Păun.

Theorem 6.3 ([DHP24, Theorem 1.1]). *Let (X, B) be a $\mathbb{Q}$ -factorial compact Kähler fourfold dlt pair such that $K_X + B \sim_{\mathbb{Q}} M \geq 0$. Then (X, B) has a log minimal model.*

Theorem 6.4 ([DHP24, Corollary 7.12]). *Let (X, B) be a $\mathbb{Q}$ -factorial compact Kähler plt pair of dimension 4 such that $\kappa(X, K_X + B) \geq 0$. Then (X, B) has a log terminal model.*

 **Idea of proof.** Their proof introduces the Θ function, which appeared in [Bir12].

6.3 Kähler BCHM

The strongest existence result is [HLX26a], Hacon–Xie proved an analog of [BCHM10] in the Kähler setting.

Theorem 6.5 ([HLX26a, Theorem 1.5]). *Let $(X, B + \beta)$ be a compact, n -dimensional, Kähler, strongly $\mathbb{Q}$ -factorial, gklt pair such that $B + \beta_X$ is big, and ω a modified Kähler form such that $K_X + B + \beta_X + \omega$ is nef. Then we can run the $K_X + B + \beta_X$ minimal model program with scaling of ω which terminates with either a good log terminal model or a Mori fiber space.*

6.4 Other existence of (good) minimal model results

Das–Hacon proved the existence of good minimal models for Kähler klt pairs of maximal Albanese dimension.

Theorem 6.6 ([DH24c, Theorem 1]). *Let (X, B) be a compact Kähler klt pair of maximal Albanese dimension. Then (X, B) has a good minimal model.*

Remark 6.7 (Comparison with the projective case). Fujino [Fuj13, Theorem 4.3] proved that projective klt pairs of maximal Albanese dimension admit good minimal models. It is instructive to compare Das–Hacon’s result with Fujino’s and identify what is new in the Kähler setting.

Recently, Huang proved the existence of good minimal models for klt Kähler pairs when the Albanese map is projective and the general fibers admit good minimal models.

Theorem 6.8 ([Hua25, Theorem 1]). *Let $a_X : X \rightarrow A$ be the Albanese morphism of a compact Kähler variety X , where (X, B) is a klt pair. Assume a_X is projective with connected fibers, and let F be the general fiber of a_X . If (F, B_F) has a good minimal model, where $B_F = B|_F$ is the restricted boundary, then (X, B) has a good minimal model.*

7 Kähler Criterion and Projectivity Criterion

Hacon–Li–Xie proved that the Kähler MMP preserves the Kähler condition in arbitrary dimension.

Theorem 7.1 ([HLX26b, Theorem 1.2]). *Let $(X, B + \beta)$ be a generalized klt pair, where $B + \beta_X$ is modified big, and X is compact, Kähler, and strongly $\mathbb{Q}$ -factorial. If $f : X \rightarrow Y$ is a flipping or divisorial contraction associated to an extremal ray Γ , then Y is Kähler.*

There are several interesting criteria for projectivity and Kählerness: Claudon–Höring developed a projectivity criterion for Kähler morphisms.

Theorem 7.2 ([CH24, Theorem 1.1]). *Let $f : X \rightarrow Y$ be a fibration (that is, a surjective morphism with connected fibres) between compact Kähler manifolds. Assume one of the following holds:*

- (1) *The natural map $f^* : H^0(Y, \Omega_Y^2) \rightarrow H^0(X, \Omega_X^2)$ is an isomorphism; or*
 - (2) *The morphism f is Moishezon, i.e. there exists a line bundle L on X that is f -big.*
- Then f is a projective morphism.*

8 Canonical Bundle Formulas

The canonical bundle formula for proper morphisms between compact Kähler manifolds remains an open question. Some partial results are known for projective morphisms between compact Kähler varieties [HP24].

Theorem 8.1 ([HP24],[HLX26a, Theorem 2.53]). *Let $f : X \rightarrow Z$ be a surjective projective morphism of normal compact Kähler varieties with connected fibers, $(X, B + \beta)$ a generalized pair which is gklt over an open subset U of Z , such that $B + \beta_X$ is modified big, $K_X + B + \beta_X \equiv f^* \gamma$ for some $\gamma \in H_{BC}^{1,1}(Z)$. Then $(Z, \Delta_Z + \delta)$ is a generalized pair such that $(U, \Delta|_U + \delta)$ is gklt, i.e.*

(1) $\Delta_Z \geq 0$,

(2) δ is b-nef, that is δ descends to some model Z' so that $\delta_{Z'}$ is nef,

(3) the coefficients of components of $\Delta_{Z'}$ whose images intersect U are < 1 for any birational model $Z' \rightarrow Z$.

Moreover, possibly after replacing $(X, B + \beta)$ by a generalized pair $(X, B' + \beta')$ where $K_X + B + \beta_X \equiv K_X + B' + \beta'_X$ and $(X, B' + \beta')$ satisfies the hypothesis of the theorem, we may assume that δ_Z is modified big.

9 Abundance Conjecture for Kähler Pairs

Abundance for lc pairs on Kähler threefolds was proved by Demailly–Ou [DO24].

Theorem 9.1 ([DO24, Theorem 1.1] + [DO25, Corollary 1.3]). *Let (X, Δ) be an lc pair such that X is a compact Kähler threefold. If $K_X + \Delta$ is nef, then it is semi-ample.*

10 Several Applications of Kähler minimal model program

10.1 Deformation of Positivity in Kähler Families

Hacon–Li–Rao proved the following deformation result for pseudo-effective adjoint classes in families, which was later generalized to Kähler families without assuming that the central fibre is projective.

Theorem 10.1 ([HLR26],[HLX26a, Corollary 7.8]). *Let $f : X \rightarrow S$ be a family from a Kähler variety X onto a smooth, connected, and relatively compact curve S . Assume that the fibers X_t have canonical singularities for all $t \in S$. Then K_{X_0} is pseudo-effective if and only if K_{X_t} are pseudo-effective for all $t \in S \setminus \{0\}$, and X_0 is uniruled if and only if X_t are uniruled for all $t \in S \setminus \{0\}$.*

The proof depend on the following deformation invariance of big and nef adjoint classes.

Theorem 10.2 ([HLR26, Proposition 4.24]). *Let $f : X \rightarrow S$ be a proper surjective morphism from a generalized Kähler pair $(X, B + \beta)$ onto a smooth, connected, relatively compact curve S . Fix a point $0 \in S$ and assume that the support of the boundary divisor B does not contain the fiber X_0 .*

Assume that the restriction to the central fiber $(X_0, B_0 + \beta_0)$ is a generalized klt pair, such that $K_{X_0} + B_0 + \beta_0$ is nef and big. Then $K_X + B + \beta_X$ is nef and big over U for some Euclidean open neighbourhood $U \subset S$ of 0 .

References

- [BCHM10] C. Birkar, P. Cascini, C.D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, Journal of the American Mathematical Society **23** (2010), no. 2, 405–468.
- [Bir12] Caucher Birkar, *Existence of log canonical flips and a special LMMP*, Publ. Math. Inst. Hautes Études Sci. **115** (2012), 325–368. MR 2929730
- [CH20] Junyan. Cao and A. Höring, *Rational curves on compact Kähler manifolds*, J. Differential Geom. **114** (2020), no. 1, 1–39.
- [CH24] Benoît Claudon and Andreas Höring, *Projectivity criteria for kähler morphisms*, 2024.
- [CHP16] F. Campana, A. Höring, and T. Peternell, *Abundance for Kähler threefolds*, Ann. Sci. Éc. Norm. Supér. (4) **49** (2016), no. 4, 971–1025.
- [DH24a] O. Das and C.D. Hacon, *On the minimal model program for Kähler 3-folds*, 2024.
- [DH24b] O. Das and C.D. Hacon, *Transcendental minimal model program for projective varieties*, 2024.
- [DH24c] Omprokash Das and Christopher Hacon, *Existence of good minimal models for Kähler varieties of maximal Albanese dimension*, C. R. Math. Acad. Sci. Paris **362** (2024), 83–91. MR 4762189
- [DH25] ———, *The log minimal model program for Kähler 3-folds*, J. Differential Geom. **130** (2025), no. 1, 151–207.
- [DHP24] O. Das, C.D. Hacon, and M. Păun, *On the 4-dimensional minimal model program for Kähler varieties*, Adv. Math. **443** (2024), Paper No. 109615, 68.
- [DHY23] O. Das, C.D. Hacon, and J.I. Yáñez, *Mmp for generalized pairs on Kähler 3-folds*, 2023.

- [DO24] O. Das and W.-H. Ou, *On the log abundance for compact Kähler threefolds*, Manuscripta Math. **173** (2024), no. 1-2, 341–404.
- [DO25] Omprokash Das and Wenhao Ou, *On the log abundance for compact Kähler threefolds ii*, 2025.
- [Fuj13] Osamu Fujino, *On maximal Albanese dimensional varieties*, Proc. Japan Acad. Ser. A Math. Sci. **89** (2013), no. 8, 92–95. MR 3127923
- [HLR26] Christopher D. Hacon, Yi Li, and Sheng Rao, *On pseudo-effectivity and volumes of adjoint classes in kähler families with projective central fiber*, 2026.
- [HLX26a] Christopher Hacon, Yi Li, and Lingyao Xie, *Fujiki class $\mathbb{J}$ varieties and a kähler criterion*, 2026.
- [HLX26b] ———, *Fujiki class $\mathbb{J}$ varieties and a kähler criterion*, 2026.
- [HP16] A. Höring and T. Peternell, *Minimal models for Kähler threefolds*, Invent. Math. **203** (2016), no. 1, 217–264.
- [HP24] C.D. Hacon and M. Păun, *On the canonical bundle formula and adjunction for generalized kaehler pairs*, 2024.
- [Hua25] Yu-Ting Huang, *On the existence of good minimal models for kähler varieties with projective albanese map*, 2025.
- [Ou25] W.-H. Ou, *A characterization of uniruled compact kähler manifolds*, 2025.
- [Tos18] Valentino Tosatti, *Kawa lecture notes on the kähler–ricci flow*, Annales de la Faculté des sciences de Toulouse : Mathématiques **27** (2018), no. 2, 285–376.